(5) Gamma Formulae

(i) $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx = (n-1)!$ (ii) $\int_0^{\infty} e^{-ax} x^n dx = \frac{n!}{a^{n+1}}$

(6) Area of the region bounded by a curve and lines

(i) The area of the region bounded by a curve, above x-axis and the lines $x = a$ and $x = b$ is $A = \int_a^b y dx$.

(ii) The area of the region bounded by a curve, below x-axis and the lines $x = a$ and $x = b$ is $A = -\int_a^b y dx = \left| \int_a^b y dx \right|$.

(iii) Thus area of the region bounded by the curve to the right of y-axis, the lines $y = c$ and $y = d$ is $A = \int_c^d x dy$.

(iv) The area of the region bounded by the curve to the left of y-axis, the lines $y = c$ and $y = d$ is $A = -\int_c^d x dy = \left| \int_c^d x dy \right|$.

(v) If $f(x) \geq g(x)$, then area bounded by the curves and the lines $x = a$, $x = b$ is

$$A = \int_a^b [f(x) - g(x)] dx = \int_a^b (y_U - y_L) dx$$

(vi) If $f(y) \geq g(y)$, then area bounded by the curves and the lines $y = c$, $y = d$ is

$$A = \int_c^d [f(y) - g(y)] dy = \int_c^d (x_R - x_L) dy$$

(7) Volume of the solid of revolution

(i) The volume of the solid of revolution about x-axis is $V = \pi \int_a^b y^2 dx$.

(ii) The volume of the solid of revolution about y-axis is $V = \pi \int_c^d x^2 dy$.

ICT CORNER

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Open the Browser, type the URL Link given (or) Scan the QR code. GeoGebra work book named "**12th Standard Mathematics Vol-2**" will open. In the left side of work book there are chapters related to your text book. Click on the chapter named "**Applications of Integration**". You can see several work sheets related to the chapter. Go through all the work sheets

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